

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO: 38. Face Detection Using OpenCV

Aim

To detect the face in **MICKEY.png** using OpenCV Haar Cascade Classifier.

Algorithm

1. Import OpenCV.
2. Read the image **MICKEY.png**.
3. Convert the image to grayscale.
4. Load the Haar Cascade face detector.
5. Detect the face.
6. Draw a rectangle around the detected face.
7. Display and save the result.

CODE:

```
EXP-38.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
)
# Find contours
contours, _ = cv2.findContours(
    binary,
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)

# Find the largest contour
if len(contours) > 0:
    largest = max(contours, key=cv2.contourArea)

    # Get bounding rectangle
    x, y, w, h = cv2.boundingRect(largest)

    # Draw rectangle
    cv2.rectangle(
        img,
        (x, y),
        (x + w, y + h),
        (0, 255, 0),
        2
    )

    # Add label
    cv2.putText(
        img,
        "Face/Object",
        (x, y - 10),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0, 255, 0),
        2
    )

    print("Face/Object detected successfully!")
else:
```

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py
Traceback (most recent call last):
  File "C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py", line 15, in <module>
    face_cascade = cv2.CascadeClassifier(
AttributeError: module 'cv2' has no attribute 'CascadeClassifier'
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-38.py
Face/Object detected successfully!
```



Ln: 4 Col: 0

Result

The face in MICKEY.png is successfully detected and marked with a rectangular bounding box using OpenCV.